

The Best Open Source Platforms *for* Developing Blockchain Applications



Since the phenomenal rise in the value of the cryptocurrency Bitcoin, which uses a blockchain for accounting, there has been a tremendous interest in blockchain technology. A blockchain is a transparent distributed ledger which is revolutionising the financial services industry by empowering millions across the globe to authenticate and transact immediately, without the involvement of expensive intermediaries. This article introduces a few popular open source platforms that help in the quick and easy development of blockchain based applications.

Blockchain technology is termed as the 'Fifth Evolution' of computing. A blockchain is regarded as a novel approach to distributed databases. It is a disseminated database of open records which documents all transactions or advanced events that have been executed prior to that point in time when it is shared among participating members. Every transaction in the public record is confirmed by agreement by a larger number of members within that framework.

These days, transactions between persons or companies are often centralised and controlled by a third party organisation. Making a digital payment or currency transfer requires a bank or credit card provider as the middleman to complete the transaction.

In addition, a transaction attracts a fee from banks or credit card companies. The same process applies to other application domains like games, software, etc. The transaction system is centralised, and all data and information are controlled and managed by a third party organisation, rather than the two principal participants in the transaction. Blockchain technology has been developed to solve this issue.

A blockchain is a data structure that makes it possible to create a digital ledger of data and share it among a network of independent parties. The ultimate goal of blockchain technology is to create a decentralised environment where no third party is in control of the data and the transactions. A blockchain is a distributed database solution

that maintains a growing list of data records that are confirmed by the nodes participating in it. The data is recorded in a public ledger, which includes all the information about every transaction ever completed. This information is shared and available to all nodes. This, in turn, makes the blockchain system more transparent and leaves no room for third parties.

Blockchains are secure by design and include sophisticated distributed computing systems with high Byzantine Fault Tolerance. Blockchain technology was invented by Satoshi Nakamoto in 2008 to serve as a public transaction ledger of the cryptocurrency called Bitcoin. The invention of the blockchain for Bitcoin made it the first digital currency to solve the double-spending problem